

REVIEW ARTICLE OPEN

Review of the sustainability of food systems and transition using the Internet of Food

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Many current food systems are unsustainable because they cause significant resource depletion and unacceptable environmental impacts. This problem is so severe, it can be argued that the food eaten today is equivalent to a fossil resource. The transition to sustainable food systems will require many changes but of particular importance will be the harnessing of internet technology, in the form of an 'Internet of Food', which offers the chance to use global resources more efficiently, to stimulate rural livelihoods, to develop systems for resilience and to facilitate responsible governance by means of computation, communication, education and trade without limits of knowledge and access. A brief analysis of the evidence of resource depletion and environmental impact associated with food production and an outline of the limitations of tools like life cycle assessment, which are used to quantify the impact of food products, indicates that the ability to combine data across the whole system from farm to human will be required in order to design sustainable food systems. Developing an Internet of Food, as a precompetitive platform on which business models can be built, much like the internet as we currently know it, will require agreed vocabularies and ontologies to be able to reason and compute across the vast amounts of data that are becoming available. The ability to compute over large amounts of data will change the way the food system is analysed and understood and will permit a transition to sustainable food systems.

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INTRODUCTION

The food we eat today is unsustainable for two reasons: the food system causes unacceptable environmental impacts and it is depleting non-renewable resources. Our food can be regarded as 'fossil food' because its production relies on fossil fuel, non-renewable mineral resources, depletion of groundwater reserves and excessive soil loss. The idea of sustainable food systems is at the heart of global efforts to manage and regulate human food supply.¹ The sustainable development goals focus on a number of critical global issues, but Goal 2 ('end hunger, achieve food security and improved nutrition and promote sustainable agriculture'), Goal 12 ('ensure sustainable consumption and production patterns') and Goal 13 ('take urgent action to combat climate change and its impacts') are intimately related to the need to transition global food systems from fossil to sustainable. To understand how to meet the challenge presented by these goals, it is necessary to consider what is meant by 'sustainable' in the context of a food system. In 1989, the Food and Agriculture Organisation (FAO) council defined sustainable development as 'the management and conservation of the natural resource base, and the orientation of technological and institutional change in such a manner as to ensure the attainment and continued satisfaction of human needs for present and future generations. Such sustainable development (in the agriculture, forestry and fisheries sectors) conserves land, water, plant and animal genetic resources, is environmentally non-degrading, technically appropriate, economically viable and socially acceptable'.² The important ideas in this definition are working within the planetary boundary ('the natural resource base'), having a future-proof

system ('continued satisfaction', 'present and future generations'), limiting impacts to those manageable by the buffering capacity of the planet ('environmentally non-degrading'), considering the financial needs of business stakeholders ('economically viable') and compatible with local needs and customs ('socially acceptable').

Five principles have been identified to support a common vision for sustainable agriculture and food.³ These are: (1) resource efficiency; (2) action to conserve, protect and enhance natural resources; (3) rural livelihood protection and social well-being; (4) enhanced resilience of people, communities and ecosystems; and (5) responsible governance. The aim of this paper is to outline the case for why food systems are not sustainable and to define the case for using technology, specifically internet technologies (hardware and software combined to make the 'Internet of Food') to enable the transition of the food system from fossil to sustainable. Increasing population, increasing consumption, a billion malnourished people and agriculture that is concurrently degrading land, water, biodiversity and climate on a global scale⁴ combine to indicate that the fossil food systems we currently rely on are not fit-for-purpose. It is suggested that halting agricultural expansion, closing yield gaps, increasing efficiency, changing diets and reducing waste could lead to a doubling of food production with reduced environmental impacts of agriculture.⁴ To achieve these changes, it is going to be necessary to harness internet technology, in the form of an 'Internet of Food', which offers the chance to use global resources more efficiently, to stimulate rural livelihoods, to develop systems for resilience and to facilitate responsible governance by means of computation, communication, education and trade without limits of knowledge and access.

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The concept of 'Internet of Food' first appeared in peer-reviewed literature in 2011 (based on a search of scopus.com using 'Internet of Food' as the search term). It was described by the idea of food items having an 'IP identify', which raised the question of how this might influence our eating habits.⁵ Their focus was very much on how the technology could influence food choices and predicted that by 2020 it would be possible to monitor and control food objects remotely through the Internet. It is this technological control of the food system that has real potential to help societal stakeholders (consumers, retailers, processors, producers, shareholders, landowners, indigenous peoples and so on) to engage in the transition of our food system from being fossil to sustainable. The ubiquitous physical tagging and sensing of mass and energy flow in the food system linked to a formal semantic web will allow computation over the whole system to answer questions such as: What was the resource depletion of this product? What is the social impact of eating this product? What food safety procedures have been employed for this product? What and where has wealth been created by the value chain of this product? When these questions can be answered for specific instances of food product types and predicted for new products, then it will be possible to determine whether a specific food system is sustainable or not. The stakeholders demanding answers to these questions are likely to be governance and policy makers and consumers. When these questions can be answered, it will be possible to plan how to manage the evolution of the fundamental life support system (food) from fossil to sustainable in order to support a growing global population.

CURRENT FOOD SYSTEMS

To understand the need for a systematic transformation of the food system, it is necessary to detail exactly why it is unsustainable. An overwhelming case can be made for the environmental dimension of the system, but there are also social and economic issues as well. This paper will focus the environmental case (resource depletion and adverse environmental impact that relate to the 'continued satisfaction' and 'environmentally non-degrading' criteria for sustainable food systems), but similar cases can be made for important social and economic issues as well.

Resource depletion

The resource depletion case can be made with respect to energy, nutrients, water, soil and land. Each will be summarised in turn. To date, the agri-food system has converted non-renewable fossil fuel energy into food by enabling mechanisation, amplified fertiliser production, improved food processing and safe global transportation.⁶ According to FAO,⁷ the agri-food sector accounts for around 30% of the world's total energy consumption, with Europe alone accounting for 17% of gross energy consumption in 2013.⁸ Agriculture, including crop cultivation and animal rearing, is the most energy-intense phase of the food system, accounting for nearly one third of the total energy consumed in the food production chain.⁹ To date, renewable energy has had limited penetration of the agri-food sector with fossil fuels accounting for almost 79% of the energy consumed by the food sector.⁸ From an energy perspective, the food system can be regarded as unsustainable (cannot meet the 'continued satisfaction' requirement) due to its reliance on fossil energy sources.

By the end of the 20th Century, it was estimated that US-produced ammonia represented 32% of fertiliser nitrogen (N) demand, which was produced by extracting N from the atmosphere as ammonia by a process using hydrogen from natural gas (fossil fuel).⁹ The vast majority of N fertilisers consumed today are still created using fossil fuels and cannot be regarded as sustainable until such times as new technological approaches

emerge, which are currently in their infancy.^{10,11} A review of mineral fertiliser reserves concluded that potash reserves (the source of most potassium (K) fertilisers) are of great concern and that it is time to start evaluating other sources of K for agriculture¹² but others concluded that 'modern agriculture is currently relying on a non-renewable resource and future phosphate rock is likely to yield lower quality P at a higher price'.¹³ If significant physical and institutional changes are not made to the way we currently use and source P, agricultural yields will be severely compromised in the future. Estimates for when world peak P will be reached range from 2027¹⁴ to 2033.¹³ Variations in estimations of when peak P will occur are due to the constant changing of reserve levels.¹² The 'power imbalance' where just three countries controlling >85% of the known global phosphorus reserves,¹⁵ a concentration of power far greater than that of crude oil, is also of concern, and it has been concluded that the combined impact increasing demand, dwindling reserves and geopolitical constraints could result in reduced production and supply of chemical P fertilisers and increased global P price.¹⁶ It is clear that over time horizons of around 50 years the agri-food system is going to face a major nutrient crisis unless reliance of fossil mineral resources is significantly reduced and ultimately eliminated. From a nutrient management point-of-view, the food system can be regarded as unsustainable (cannot meet the 'continued satisfaction' requirement) due to its reliance on fossil mineral resources.

Modern food production is reliant on irrigation to a great extent, which according to the UN water programme, accounts for 70% of freshwater withdrawals worldwide.¹⁷ Excessive removal of groundwater for irrigation is leading to rapid depletion of aquifers in key food-producing regions, such as North-Western India, the North China Plain, Central USA and California.¹⁸ Aquifers replenish so slowly that they are effectively a non-renewable resource. The depletion of these large freshwater stocks threatens food production and security locally and globally via international food trade. Non-sustainable groundwater abstraction contributed to 20% of global gross irrigation water demand in the year 2000 with this demand having tripled over the period 1960–2000.¹⁹ For many countries, irrigation is sustained by non-renewable groundwater, and it has been highlighted that 'a vast majority of the world's population lives in countries sourcing nearly all their staple crop imports from partners who deplete groundwater to produce these crops'.¹⁸ Countries who both produce and import food irrigated from rapidly depleting aquifers are particularly at risk, such as USA, Mexico, Iran and China. It has been estimated that India, soon to be the most populous country in the world, will be unable to meet water requirements within 300 years and emerging pressures may reduce this time horizon considerably.²⁰ Given the interaction of water supply with energy, this situation may become even worse. For example, in California, 20% of electricity production is used for moving and pumping water for agriculture,²¹ and as water becomes more difficult to access, the energy demand will increase. From a water management point-of-view, the food system can be regarded as unsustainable (cannot meet the 'continued satisfaction' requirement) due to its reliance on non-renewable water resources.

Over 20 years ago, it was estimated that around one third of the world's agricultural land had been lost to erosion and the rate of loss was about 10 Mha/year.²² Calculations suggest that soil erosion rates under ploughed cultivation are one to two orders of magnitude greater than soil production rates.²³ This rate of soil loss is not compatible with the 'continued satisfaction' requirement for a sustainable food system. It is also linked with other environmental impacts, such as loss of carbon, gaseous emissions, non-point source pollution and sedimentation of waterways,²⁴ therefore it is not compatible with the 'environmentally non-degrading' criteria as well. Given projections for expansion of dryland areas to around 50% of total land surface, with 78% of